

PROJECT COMPLETION REPORT

1. Project Title :

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

An End-to-End Data Analytics Project Using Python, SQL, and Power BI

2. Project Objective :

The primary objective of this project is to analyze customer shopping behavior and identify meaningful trends in purchasing patterns, customer segments, product preferences, and subscription behavior using data analytics techniques.

- To clean and preprocess the customer shopping dataset using **Python and Pandas**.
- To perform **exploratory data analysis (EDA)** and understand customer purchasing patterns.
- To use **SQL/PostgreSQL** to analyze revenue, discounts, products, shipping, subscriptions, and customer loyalty.
- To segment customers into **New, Returning, and Loyal** groups based on their purchase history.
- To develop an interactive **Power BI dashboard** for visualizing important business insights.
- To identify factors influencing **customer engagement, repeat purchases, and spending behavior**.
- To provide **actionable business recommendations** that can help improve sales, customer satisfaction, loyalty, and marketing strategies.

3. Project Description in detail:

3.1 Overview :

The Customer Shopping Behavior Analysis project is an end-to-end data analytics project focused on understanding customer purchasing patterns and generating actionable business insights. The project uses transactional data containing 3,900 purchases and 18 columns, covering customer demographics, purchase details, shopping behavior, reviews, and shipping information.

The project follows a structured analytics workflow starting with data preparation and exploratory analysis using Python. The cleaned data is then integrated with PostgreSQL for SQL-based business analysis. Finally, Power BI is used to create an interactive dashboard that presents important trends and insights in a visual format.

The main objective is to understand customer spending behavior, product preferences, discount usage, subscription behavior, customer loyalty, and revenue contribution across different customer groups.

3.2 Project Approach

The project was completed through the following seven steps:

Step 1 – Business Problem Identification

The first step was to understand the business problem and identify the key questions that needed to be answered. The focus was on understanding customer behavior and using the available data to improve customer engagement, sales, loyalty, and marketing strategies.

Step 2 – Data Preparation and EDA using Python

The raw customer shopping dataset was imported into Python using Pandas. Initial exploration was performed using data structure checks and descriptive statistics. Missing values were identified and the missing review ratings were handled using the median rating of the respective product category.

Additional preprocessing included column standardization and feature engineering. An `age_group` column and `purchase_frequency_days` feature were created to support further analysis.

Step 3 – Database Integration and SQL Analysis

After preprocessing, the cleaned dataset was loaded into PostgreSQL for structured analysis. SQL queries were written to answer important business questions related to revenue, products, discounts, shipping, subscriptions, customer segmentation, and purchasing behavior.

A total of **10 business questions** were analyzed using SQL, including gender-wise revenue, high-spending discount users, top-rated products, shipping comparison, subscription behavior, customer segmentation, and revenue by age group.

Step 4 – Power BI Dashboard Development

The analyzed data was connected to Power BI to create an interactive dashboard. The dashboard was designed to provide a visual understanding of customer behavior and important business metrics.

Step 5 – Insight Generation

The results from SQL analysis and Power BI visualizations were examined to identify important patterns in customer spending, product performance, subscription behavior, shipping preferences, and customer loyalty.

Step 6 – Business Recommendations

Based on the identified insights, recommendations were developed around subscription promotion, loyalty programs, discount optimization, product positioning, and targeted marketing.

Step 7 – Project Documentation

The final stage involved documenting the methodology, analysis, findings, and recommendations in the project report and presentation. The project files can also be organized into a repository containing the Python notebook, SQL queries, dataset, and Power BI dashboard.

3.3 Tools and Technologies Used :

Technology	Purpose
Python	Data preprocessing and exploratory data analysis
Pandas	Data loading, cleaning and transformation
Jupyter Notebook	Executing and documenting Python analysis
PostgreSQL	Storing and analyzing the cleaned dataset
SQL	Performing business analysis and extracting insights
Power Bi	Creating interactive dashboards and visualizations
GitHub	Organizing and sharing project files

The overall workflow follows the requested deliverables of Python-based data preparation, SQL analysis, Power BI visualization, and project documentation.

3.4 Insights from the Dashboard :

The Power BI dashboard provides a visual summary of the major findings from the analysis.

Key Insights

- **Revenue by Gender:** Female customers generate slightly higher total revenue than male customers.

- **Shipping Behavior:** Customers using Express Shipping have a higher average purchase amount than Standard Shipping customers.
- **Subscription Behavior:** Subscription status shows a meaningful relationship with customer spending and revenue contribution.
- **Customer Segmentation:** Customers are classified into New, Returning, and Loyal segments based on their purchase history.
- **Product Performance:** The analysis identifies products with high average customer ratings and the most purchased products within categories.
- **Discount Behavior:** Certain products have a higher proportion of purchases made using discounts.
- **Age-Group Revenue:** Revenue contribution varies across different customer age groups.

These insights can help management understand customer behavior and identify areas where marketing, loyalty, subscription, and product strategies can be improved.

3.5 Real-World Impact for Retail Business and Customer Engagement :

This project can provide practical value to a retail business by helping decision-makers understand how customers interact with products, discounts, subscriptions, and shipping options.

The analysis can support businesses in:

- **Improving customer retention** through loyalty programs.
- **Increasing subscription adoption** by offering exclusive benefits.
- **Optimizing discount strategies** while balancing customer attraction and revenue.
- **Improving product marketing** by promoting highly rated and popular products.
- **Developing targeted marketing campaigns** for high-revenue customer groups.
- **Understanding customer segments** and designing suitable strategies for New, Returning, and Loyal customers.

The reference business problem specifically focuses on improving sales, customer satisfaction, and long-term loyalty through customer shopping data.

4. Dataset Summary :

The dataset used in this project contains customer shopping transaction data collected across different product categories. It provides information about customer demographics, purchase details, and shopping behavior. The dataset was used to identify patterns in customer spending, product preferences, discounts, shipping choices, subscriptions, and customer loyalty.

4.1 Dataset Statistics :

Attribute	Details
Total Records	3,900
Total Columns	18
Dataset Type	Customer Shopping / Transaction Data
Main Focus	Customer purchasing behavior
Missing Values	37
Column with Missing Values	Review Rating

The dataset contains 3,900 purchases and 18 attributes. The 18 attributes cover customer demographics, purchase information, and shopping behavior.

4.2 Major Data Categories

1. Customer Demographics

- Age
- Gender
- Location
- Subscription Status

2. Purchase Details

- Item Purchased
- Category
- Purchase Amount
- Season
- Size
- Color

3. Shopping Behavior

- Discount Applied
- Promo Code Used
- Previous Purchases
- Frequency of Purchases
- Review Rating
- Shipping Type

These attributes provide the information required to analyze customer purchasing patterns and business performance.

4.3 Data Quality

During the initial data exploration, 37 missing values were identified in the **Review Rating** column. These missing values were handled during the Python preprocessing stage by replacing them with the median review rating of the corresponding product category.

The dataset was subsequently standardized and additional features such as **age_group** and **purchase_frequency_days** were created to support deeper analysis.

4.4 Purpose of the Dataset

The prepared dataset serves as the foundation for the complete project workflow:

Customer Data → Python Preprocessing → PostgreSQL → SQL Analysis → Power BI Dashboard → Business Insights

It enables the project to investigate revenue patterns, customer segmentation, product ratings, discount behavior, shipping preferences, subscription behavior, repeat purchases, and age-group revenue contribution.

5. Project Execution Details :



Step 1 – Business Problem Identification

The first stage of the project was to understand the business problem and define the objectives of the analysis. The retail company wanted to understand customer shopping behavior in order to improve sales, customer satisfaction, and long-term customer loyalty.

The analysis focused on identifying purchasing patterns across customer demographics and product categories. The project also examined the impact of factors such as discounts, reviews, seasons, shipping preferences, subscriptions, and previous purchases on customer behavior.

The main business question addressed in this project was:

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

Based on this business problem, a set of analytical questions was defined for further investigation.

Step 2 – Data Preparation and Exploratory Data Analysis using Python

After defining the business problem, the customer shopping dataset was imported into Python using the **Pandas** library.

The dataset contains **3,900 purchase records and 18 columns** covering customer demographics, purchase information, and shopping behavior. Important attributes include age, gender, location, subscription status, item purchased, category, purchase amount, discount information, previous purchases, review rating, and shipping type.

Data Exploration

Initial exploration was performed to understand the structure and characteristics of the dataset. Functions such as `df.info()` and `df.describe()` were used to examine:

- Number of records and columns
- Data types
- Statistical summaries
- Missing values
- Numerical and categorical attributes

Missing Value Treatment

During the data-quality check, 37 missing values were identified in the Review Rating column.

Instead of removing those records, the missing review ratings were imputed using the **median rating of the respective product category**. This helped retain the available transaction records while providing a reasonable replacement for missing values.

Column Standardization

The column names were standardized into **snake_case format** to improve readability, consistency, and ease of use during SQL analysis.

Feature Engineering

Additional features were created to support business analysis:

- **Age Group:** Customer ages were grouped into meaningful age categories.
- **Purchase Frequency:** A `purchase_frequency_days` feature was created from the purchase-frequency information.

These derived features helped make the customer behavior analysis more meaningful.

Data Consistency Check

The relationship between `discount_applied` and `promo_code_used` was examined to identify redundant information. Based on this consistency check, `promo_code_used` was removed from the cleaned dataset.

Step 3 – PostgreSQL Database Integration

Once the preprocessing was completed, the cleaned DataFrame was connected to **PostgreSQL**.

The purpose of this step was to move the prepared data into a structured relational database environment where SQL could be used for business analysis.

The cleaned dataset was loaded into PostgreSQL and made available through a customer table for executing analytical queries.

This stage created the connection between the Python preprocessing layer and the SQL analysis layer:

Python → Cleaned Data → PostgreSQL → SQL Analysis

Step 4 – Data Analysis using SQL

After loading the cleaned data into PostgreSQL, SQL queries were developed to answer the defined business questions.

A total of 10 business questions were analyzed.

Q1. Revenue by Gender

The first query compares the total revenue generated by male and female customers using `SUM(purchase_amount)` and grouping the results by gender.

Q2. High-Spending Discount Users

The second analysis identifies customers who used discounts but still spent at or above the overall average purchase amount. A subquery is used to calculate the average purchase amount.

Q3. Top 5 Products by Average Rating

The third query calculates the average review rating for each product and returns the five products with the highest average ratings.

Q4. Standard vs. Express Shipping

The fourth analysis compares the average purchase amount between Standard and Express shipping types.

Q5. Subscribers vs. Non-Subscribers

The fifth query compares subscribers and non-subscribers based on:

- Number of customers
- Average purchase amount
- Total revenue

This helps evaluate the relationship between subscription status and customer spending.

Q6. Discount-Dependent Products

The sixth query calculates the percentage of purchases where discounts were applied for each product and identifies the top five products with the highest discount rate.

Q7. Customer Segmentation

Customers were classified into three segments according to their previous purchase count.

A **CASE** statement was used to create the customer segment, followed by a count of customers in each segment.

Q8. Top 3 Products within Each Category

A window function using **ROW_NUMBER()** was used to rank products within each category according to the number of orders. The top three products from every category were then selected.

Q9. Repeat Buyers and Subscription

The ninth analysis focuses on customers with more than five previous purchases and compares their subscription status. This helps understand whether repeat purchasing behavior is associated with subscription adoption.

Q10. Revenue by Age Group

The final query calculates the total revenue generated by each age group and orders the groups according to their revenue contribution.

Step 5 – Power BI Dashboard Development

After completing the SQL analysis, the data and analytical results were used to create an interactive dashboard in **Microsoft Power BI**.

The purpose of the dashboard was to transform the analytical results into an easy-to-understand visual format for business stakeholders.

The dashboard presents important aspects of customer shopping behavior, including:

- Revenue performance
- Gender-wise revenue
- Product ratings
- Shipping behavior
- Subscription performance
- Customer segmentation
- Product preferences
- Customer purchasing patterns

The dashboard allows users to understand important trends through visual charts, KPI-style metrics, and interactive analysis.

Step 6 – Insight Generation and Business Recommendations

The results obtained from SQL analysis and Power BI visualization were interpreted to identify meaningful business insights.

Some important findings include:

- Female customers generated slightly higher total revenue than male customers.
- Express-shipping customers showed a higher average purchase amount.
- Some products received consistently high customer ratings.
- Customer behavior can be divided into New, Returning, and Loyal segments.
- Subscription behavior provides an opportunity for improving customer engagement.
- Certain products have a higher dependence on discounts.

The analysis was then converted into practical recommendations. The project recommends strengthening subscription benefits, developing loyalty programs, reviewing discount strategies, promoting highly rated products, and using targeted marketing for valuable customer groups.

Step 7 – Project Documentation and Presentation

The final stage was to communicate the project findings clearly through documentation and presentation.

The project report documents the complete process, including:

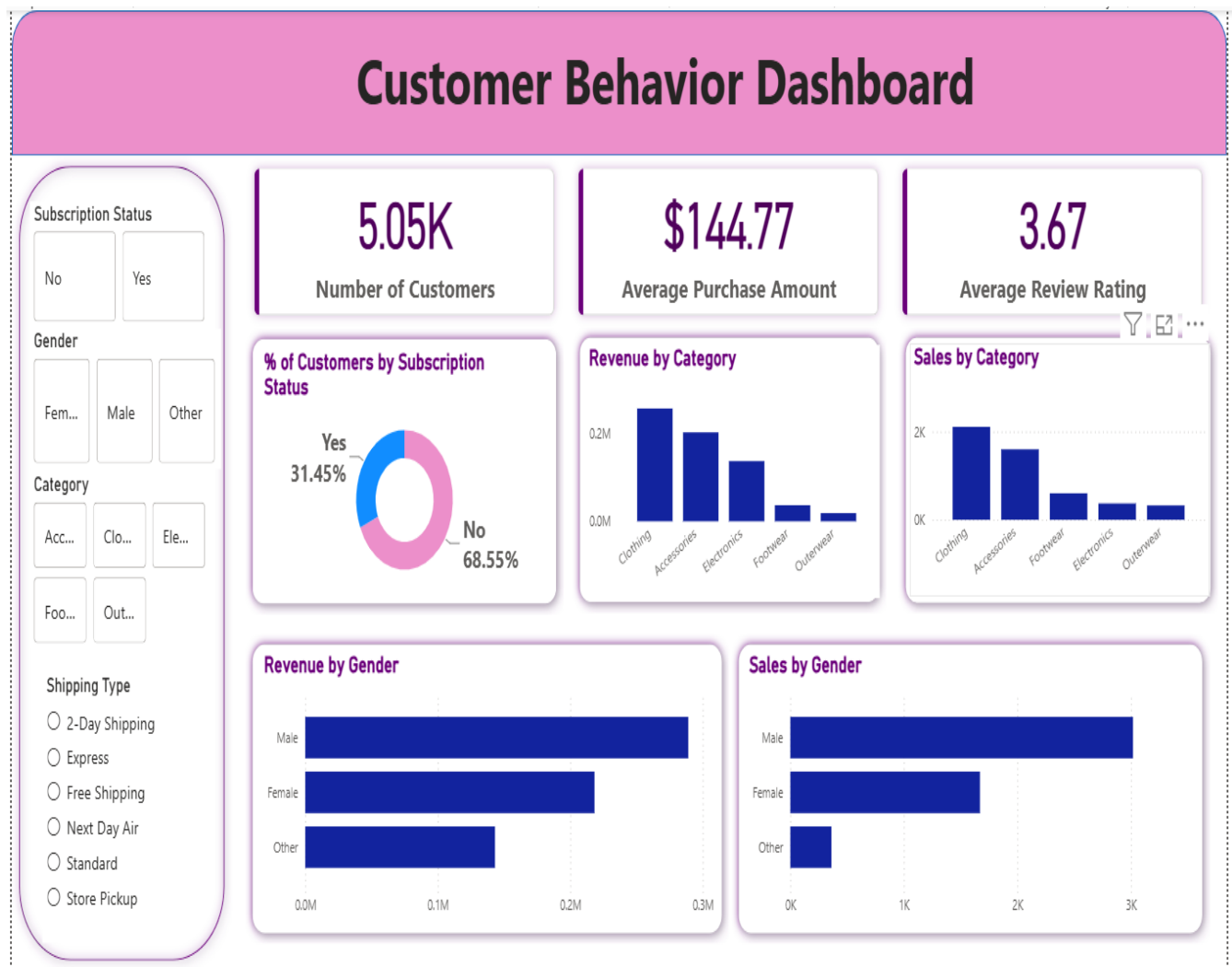
Business Problem → Data Preparation → SQL Analysis → Power BI Dashboard → Insights → Recommendations

A presentation was also prepared to communicate the major findings and recommendations in a concise and visual format.

The project deliverables include the data preparation and analysis workflow, SQL queries, Power BI dashboard, report, and presentation. The business-problem document specifically identifies these as the expected project deliverables.

6. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



7. Business Recommendations :

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.

8. Conclusion :

The Customer Shopping Behavior Analysis project successfully analyzed customer purchasing patterns using an end-to-end data analytics approach involving Python, PostgreSQL/SQL, and Power BI. The dataset consisted of 3,900 purchase records across 18 attributes, providing information about customer demographics, product preferences, purchasing behavior, discounts, subscriptions, shipping, and reviews.

Python was used for data loading, exploration, cleaning, missing-value handling, and feature engineering. The cleaned data was then integrated with PostgreSQL, where SQL queries were used to answer important business questions related to revenue, customer segments, product performance, discounts, subscriptions, and purchasing behavior.

The results were presented through an interactive Power BI dashboard, making the analytical findings easier to understand and use for decision-making. The analysis highlighted opportunities related to subscription growth, customer loyalty, targeted marketing, discount strategies, and product positioning.

Overall, this project demonstrates how raw customer data can be transformed into meaningful business insights through a structured analytics workflow. The findings can help a retail business better understand its customers, improve engagement, optimize marketing strategies, and support data-driven business decisions.